

Title

Community-centered approach to reversing metabolic syndrome: 18-week results of an integrated program combining exercise, education, and clinical support

Abstract

Community-based structured programs may effectively address metabolic syndrome, though Latin American evidence remains limited. We evaluated an 18-week integrated program combining three weekly supervised aerobic sessions, monthly health-education workshops, and biweekly clinical monitoring in 80 adults with metabolic syndrome. Participants were randomly assigned (intervention n=33, control n=47) and groups were balanced at baseline. Intervention participants attended supervised sessions and educational activities regularly; controls received customary care. Primary outcomes measured changes in fitness capacity, body measurements, fasting glucose, lipid panel, and blood pressure. The intervention group demonstrated greater gains in fitness capacity, reductions in waist circumference and body weight, lower fasting glucose and triglycerides, improved blood pressure, and favorable shifts in cholesterol fractions. Within the intervention group, significant improvements in fitness, weight loss, fat mass reduction, and lean muscle gain were evident. These results demonstrate that an integrated, accessible community program can produce meaningful improvements across multiple cardiometabolic domains in adults with metabolic syndrome.